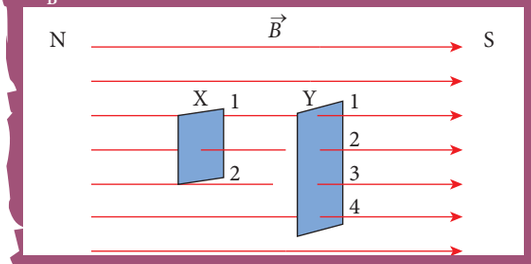
Note

Here the integral is taken over area.

Let X and Y be two planar strips whose orientation is such that the direction of area vector of planar strips is parallel to the direction of the magnetic field $\vec{B}$ as shown in figure. The number of magnetic field lines passing through area of the strip X is two. Therefore, the flux passing through area X is $\Phi_B = 2$ Wb. Similarly, the number of magnetic field lines passing through area of strip Y is $\Phi_B = 4$ Wb.



(e) Uniform magnetic field and Non-uniform magnetic field

Uniform magnetic field

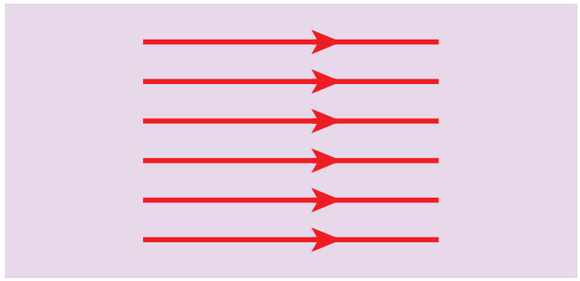


Figure 3.12 Uniform magnetic field

Magnetic field is said to be uniform if it has same magnitude and direction at all the points in a given region. Example, locally Earth's magnetic field is uniform.

The magnetic field of Earth has same value over the entire area of your school!

Non-uniform magnetic field

Magnetic field is said to be non-uniform if the magnitude or direction or both varies at all its points. Example: magnetic field of a bar magnet

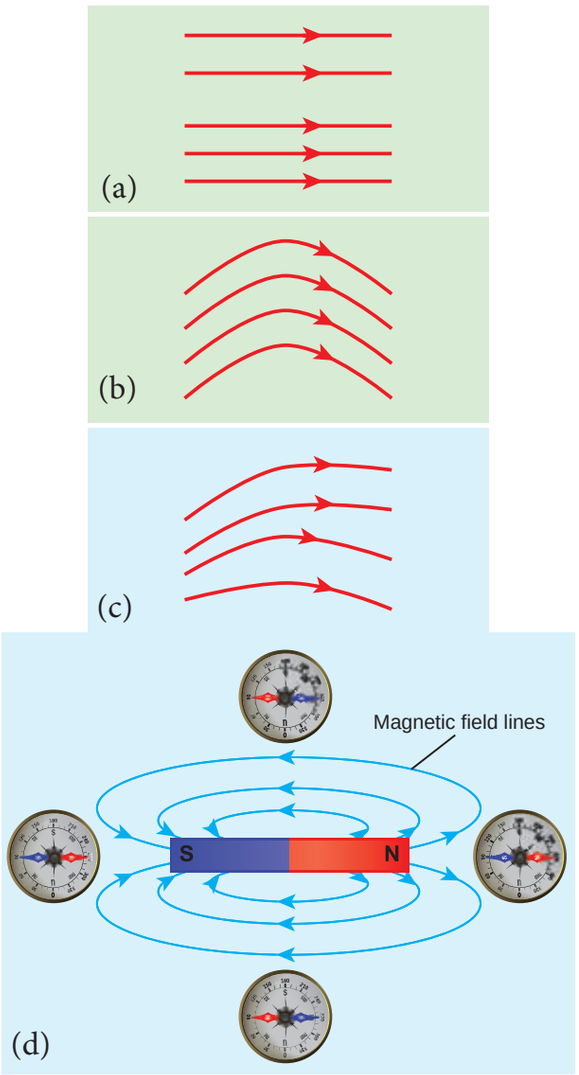


Figure 3.13 Non-uniform magnetic field – (a) direction is constant (b) direction is not a constant (c) both magnitude and direction are not constant (d) magnetic field of a bar magnet

EXAMPLE 3.4

Calculate the magnetic flux coming out from the surface containing magnetic dipole (say, a bar magnet) as shown in figure.

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